POLYCET Maths Important Formulas (Formulas Only)

1. Algebra

$(a + b)^2 = a^2 + 2ab + b^2$

$(a - b)^2 = a^2 - 2ab + b^2$

$a^2 - b^2 = (a + b)(a - b)$

$(x + a)(x + b) = x^2 + (a + b)x + ab$

Quadratic Equation: $ax^2 + bx + c = 0$

Roots: $x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$

Sum of roots: $\alpha + \beta = -b/a$

Product of roots: $\alpha\beta = c/a$

Arithmetic Progression (AP): n-th term: $a_n = a + (n-1)d$

Sum of n terms of AP: $S_n = n/2\ [2a + (n-1)d]$

Geometric Progression (GP): n-th term: $a_n = a * r^{(n-1)}$

Sum of n terms of GP: $S_n = a(r^n - 1)/(r - 1),\ r \neq 1$

Sum of infinite GP: $S_\infty = a / (1 - r), |r| < 1$

2. Trigonometry

$\sin^2\theta + \cos^2\theta = 1$

$1 + \tan^2\theta = \sec^2\theta$

$1 + \cot^2\theta = \csc^2\theta$

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$

$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$

$\tan(A \pm B) = (\tan A \pm \tan B)/(1 \mp \tan A \tan B)$

$\sin 2A = 2\sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$

$\tan 2A = 2\tan A / (1 - \tan^2 A)$

3. Geometry

Triangle Area (Heron): $A = \sqrt{[s(s-a)(s-b)(s-c)]}$, $s = (a+b+c)/2$

Triangle Area: (1/2)baseheight

Equilateral triangle area: $(\sqrt{3} /4) * a^2$

Right triangle Pythagoras: $c^2 = a^2 + b^2$

Circle: Circumference = $2\pi r$, Area = πr^2

4. Coordinate Geometry

Distance between (x_1,y_1) & (x_2,y_2): $\sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$

Midpoint: $((x_1+x_2)/2, (y_1+y_2)/2)$

Section formula (internal): $((mx_2+nx_1)/(m+n), (my_2+ny_1)/(m+n))$

Slope of line: $m = (y_2-y_1)/(x_2-x_1)$

Line equation (slope form): $y - y_1 = m(x - x_1)$

Line equation (two points): $(y - y_1)/(y_2-y_1) = (x - x_1)/(x_2-x_1)$

5. Mensuration

Cube:

Surface Area: $6a^2$

Volume: a^3

Cuboid:

Surface Area: $2(lb + bh + hl)$

Volume: lbh

Sphere:

Surface Area: $4\pi r^2$

Volume: $(4/3)\pi r^3$

Hemisphere:

Surface Area: $3\pi r^2$

Volume: $(2/3)\pi r^3$

Cylinder:

Surface Area: $2\pi r(h+r)$

Volume: $\pi r^2 h$

Cone:

Surface Area: $\pi r(l+r)$, $l = \sqrt{(r^2 + h^2)}$

Volume: $(1/3)\pi r^2 h$

6. Probability & Statistics

Probability: $P(E)$ = Number of favorable outcomes / Total outcomes

Mean: $\mu = (\Sigma x_i)/n$

Median: Middle value of ordered data

Mode: Most frequent value

Variance: $\sigma^2 = \Sigma(x_i - \mu)^2 / n$

Standard Deviation: $\sigma = \sqrt{\sigma^2}$